WEKA Classification Assignment 01

Master's level | Individual work | 5 marks | 60–90 minutes

Use WEKA Explorer to classify Iris flowers and compare two machine learning models. Your goal is to run a fair evaluation and explain what the results show.

Dataset and setup

Use iris.arff from the WEKA data folder. If unavailable, download it from the official repository below. Open Explorer → Preprocess → Open file, and set the last attribute (class) as the target. Record your WEKA version.

Tasks and marks

1. Explore the dataset — 1 marks
Report the number of instances, predictor attributes, class labels, class counts, and missing values. Inspect a plot of two predictors and describe one visible pattern.

2. Train and evaluate — 1 marks
In Classify, run trees → J48 and bayes → NaiveBayes using default model settings. For both runs, select 10-fold cross-validation and use random seed 1 under More options. Keep all attributes and apply no filters. Record the model options and complete the table below.

3. Interpret the output — 1 marks
Choose one model's cross-validation confusion matrix. Identify the most confused class pair and the number of errors in each direction; if none, state this. Explain one J48 decision rule from the displayed tree and note that the displayed full-data tree is separate from the models fitted within cross-validation.

4. Critically compare — 2 marks
In 150–200 words, recommend one model using accuracy, weighted F1, and interpretability. Explain why evaluating on training data can overestimate performance and why one small dataset with a single cross-validation run cannot establish that a model is universally better.

Measure	J48	NaiveBayes
Accuracy (%)	_____	_____
Weighted average F1 score	_____	_____

Submission

Submit one PDF report of at most 3 pages, including your name and student ID, the completed table, your answers, and three readable screenshots: dataset summary and the evaluation output for each model. Report your own results. Marks reward correct method and interpretation, not the highest accuracy.

Resources: Iris data | WEKA guide